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DBCLS BioHackathon 2026 report: Mass spectrometry data analysis workflow and visualization towards trans-omics research

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Abstract

Public repositories for mass spectrometry (MS) data contain rapidly growing collections of proteomic, metaproteomic, and metabolomic data, but differences in analytical procedures, metadata, data formats, and visualization environments limit their reuse and integration. During BioHackathon 2026, we developed complementary components of an infrastructure for standardized reanalysis for proteomics and trans-omics applications. First, we redesigned the jPOST proteomics reanalysis workflow to improve scalability, reduce manual intervention, and facilitate consistent transfer of data and metadata among the repository, reanalysis system, and jPOSTdb. The workflow uses detailed Journal of Proteome Data and Methods (JPDM) metadata and integrates results from three search engines using UniScore. Second, we developed metaproteome visualizations that connect spectral evidence with metabolic pathways and microbial phyla, enabling functional comparison among samples and experimental groups. Third, we enhanced Mass++4, a cross-platform spectrum and chromatogram viewer, by improving its connectivity with public repositories and expanding functions relevant to the inspection of proteomic and metabolomic data. Finally, we investigated the interoperability of MB-POST with the MetabolomicsHub common data model and mzTab-M. This analysis identified requirements for controlled-vocabulary mapping, explicit representation of experimental factors and sample groups, and standardization of quantitative-result formats. Together, these developments connect raw-data preservation, standardized metadata, reproducible reanalysis, biological visualization, and cross-repository interoperability. Although further implementation and large-scale validation are required, they provide a practical foundation for integrating proteomic, metaproteomic, and metabolomic data in future trans-omics studies.

Introduction

Mass spectrometry (MS) (Smith et al., 2014)-based metabolomics (Kell & Oliver, 2016) (Oliver et al., 1998) and proteomics (Lottspeich, 2018) (Shuken, 2023) (Wilkins et al., 1996) have become indispensable approaches for characterizing the molecular states of biological systems. Advances in high-resolution MS, chromatographic separation, data-dependent acquisition, data-independent acquisition, and computational analysis now enable the detection and quantification of thousands of metabolites and proteins across diverse sample types. Metabolomics provides a sensitive readout of biochemical activity and physiological responses, whereas proteomics captures changes in protein abundance, modification, localization, and interaction that cannot be inferred directly from genomic or transcriptomic data. In parallel, metaproteomics (Wilmes

et al., 2015) (Wilmes & Bond, 2004)—the large-scale analysis of proteins expressed by multiple organisms, especially microbial communities—has emerged as an approach for linking community composition to functional activity in complex ecosystems, including the human microbiome and environmental samples. By identifying both microbial taxa and their expressed functions, metaproteomics complements metagenomic and metatranscriptomic analyses and provides more direct validity of the biological processes occurring within a community.

Despite this progress, the reuse and integration of MS-based omics data remain challenging. Public repositories contain a rapidly growing collection of proteomics including metaproteomics, and metabolomics datasets, but differences in metadata completeness, experimental design, acquisition methods, identification criteria, quantitative procedures, and database versions often hinder systematic reanalysis and cross-study comparison. In particular, metaproteomic data pose additional difficulties because peptide sequences may be shared among multiple taxa, reference databases are highly complex, and protein functions must be interpreted together with their microbial origins. Furthermore, analytical results are commonly represented in each omics science-specific format and visualized separately for individual omics layers, making it difficult to investigate coordinated relationships among metabolites, proteins, microorganisms, pathways, and phenotypes. Standardized and reproducible reanalysis workflows for large-scale analyses, together with interactive visualization tools that organize results using common biological entities and pathways, are therefore required to unlock the value of publicly available MS data. Such an infrastructure would provide a foundation for trans-omics analysis (Yugi et al., 2016), in which molecular measurements from different omics layers are integrated to reconstruct the flow of biological information and generate mechanistic hypotheses about cellular and microbial systems.

To facilitate the systematic reuse of public proteomics data, the jPOST project has been developing jPOSTdb (Moriya et al., 2019), a secondary database that stores datasets reprocessed according to standardized analysis protocols. In this study, we report the work conducted during BioHackathon 2026 to establish reproducible workflows for proteomic data reanalysis and to develop tools for visualizing the resulting data, with a particular focus on metaproteomic datasets. We also describe developments aimed at extending this framework toward trans-omics analysis integrating proteomic and metabolomic information, including functional enhancements to Mass++ (Tanaka et al., 2014), a high-performance mass-spectrometry data viewer, and efforts to improve data interoperability among proteomics/metabolomics repositories. Together, these developments provide a foundation for the standardized reanalysis, visualization, and cross-omics integration of MS-based datasets.

Establishment of the jPOST reanalysis workflow

Background: Why re-analysis?

Reanalysis is particularly important for mass spectrometry-based omics, especially proteomics, for two principal reasons. The first is the potentially limited detection scope of the original analysis. Proteomic experiments and their associated database searches are generally optimized for a specific scientific objective and therefore explore only restricted aspects of the information contained in the acquired spectra. For example, a search that considers only unmodified peptides will fail to detect spectra derived from modified forms, whereas including plausible post-translational or chemical modifications can increase peptide and protein coverage. Likewise, experiments using multiple isotopic or isobaric labels may require separate searches designed to quantify particular label combinations; although appropriate for the original study, such strategies may leave other identifiable peptides unexplored. Reanalysis with a broader or differently defined search space can therefore recover additional information from the same raw data. Moreover, search engines, spectral-processing algorithms, reference databases, and statistical models are continually improving. The predominant analysis platforms have consequently changed over time—from workflows centered on tools such as Mascot (Perkins

et al., 1999) to those based on MaxQuant/Andromeda (Cox & Mann, 2008) (Cox et al., 2011) and, more recently, MSFragger (Kong et al., 2017) and the FragPipe ecosystem. Reprocessing legacy datasets using current algorithms and updated sequence databases can yield more sensitive and reliable identifications than were possible when the data were originally generated. For secondary databases in particular, systematic reanalysis independent of the original study-specific settings is therefore essential for maximizing protein coverage and producing consistently processed results across datasets.

The second reason concerns the control of false discoveries. In the context of nucleotide sequencing, the final sequence is generally treated as a discrete outcome rather than a probabilistic one. Although the raw output generated by a sequencer is inherently probabilistic, increasing sequencing coverage allows downstream analyses to treat the resulting sequence as sufficiently well determined. Consequently, this process is commonly referred to as determination or identification. In biochemistry, by contrast, the term "identification" traditionally implies the establishment of a reproducible method by which a specific component can be isolated and characterized. In mass spectrometry, assignment of a molecular species to an experimentally observed spectrum is more appropriately regarded as annotation rather than identification, because it involves searching a database and selecting the candidate that best matches the observed spectrum. In proteomics in particular, the number of possible peptide and protein sequences is extremely large, making statistical evaluation and multiple-testing consideration essential. As a result, the assignment of a specific amino acid sequence or protein to an observed spectrum is inherently probabilistic.

In omics analyses, multiple-testing correction commonly relies on the False Discovery Rate (FDR) (Benjamini & Hochberg, 1995), rather than on procedures such as the Bonferroni correction, which can be overly conservative. The False Discovery Proportion (FDP), defined as the proportion of false positives among all positive discoveries, is the quantity of direct interest for a particular set of results (Burger, 2018). However, the FDP cannot be determined directly because the identities of the false-positive results are unknown. The FDR is defined as the expected value of the FDP across repeated experiments. In conventional hypothesis-testing frameworks, the Benjamini-Hochberg (BH) procedure is widely used to control the FDR on the basis of p-values (Benjamini & Hochberg, 1995). Under the null hypothesis, properly calibrated p-values are expected to follow a uniform distribution, enabling statistical control of false discoveries across large numbers of tests.

In proteomics, however, false positives are not uniformly distributed with respect to database-search scores; rather, their frequency generally increases as the score decreases. Consequently, the Benjamini–Hochberg (BH) procedure cannot be directly applied to sequence assignment in proteomic database searches. Proteomics commonly addresses this problem using a target–decoy strategy (Elias & Gygi, 2007) (Käll et al., 2008), in which experimental spectra are searched against both biologically plausible target sequences and artificial decoy sequences, typically generated by reversing or shuffling the target protein sequences. Matches to decoy sequences provide an empirical means of estimating the number of incorrect target matches under a particular combination of spectra, search database, scoring method, and search parameters.

Unlike the BH procedure, which controls the expected FDR on the basis of p-values, the target–decoy strategy estimates the number of false-positive identifications directly from decoy matches. In proteomics, the false discovery rate (FDR) is then estimated as the FDP, the proportion of false-positive identifications among the accepted identifications, using this estimated number of false positives. Thus, the estimated FDR is inherently conditional on the dataset and analytical workflow from which it is derived.

Consequently, simply merging sets of results that were each reported at an FDR of 1%, for example, does not guarantee that the combined collection will retain an FDR of 1%, particularly when the datasets differ in composition, search space, scoring scheme, or score distribution. When results from multiple studies are integrated, the underlying spectra should therefore

ideally be re-searched together—or, at a minimum, statistically re-evaluated as a combined dataset—followed by renewed target–decoy-based FDR estimation at the relevant levels, such as the peptide-spectrum match (PSM), peptide, and protein levels. This dependence of error control on the complete search context explains why preserving only published identification tables is insufficient. Long-term storage of raw MS data, detailed metadata, and analysis parameters is indispensable for reliable data integration, and reanalysis is a prerequisite for constructing large-scale proteomic and metaproteomic resources suitable for comparative and trans-omics studies.

As described above, reanalysis is essential for maximizing the value of proteomics data. In the jPOST project, selected datasets submitted to the jPOST repository (Okuda et al., 2017) are reprocessed using a standardized strategy designed not only to reproduce the original analysis but also to provide additional value through improved identification reliability, comprehensive metadata utilization, and integration into jPOSTdb (Okuda et al., 2025). To improve the reliability and sensitivity of peptide and protein identification, we developed UniScore, a unified scoring scheme intended to reduce false-positive identifications (Tabata et al., 2025). Each dataset is searched using three different search engines, and their outputs are integrated to minimize engine-specific false negatives. Identifications from the individual engines are ranked and consolidated on the basis of UniScore, thereby enabling the complementary results of multiple search algorithms to be incorporated within a common statistical framework.

The reanalysis conditions are determined using metadata represented in the JPDM format, which is provided by Journal of Proteome Data and Methods (JPDM) (Ishihama, 2019) and substantially compatible with the Sample and Data Relationship Format (SDRF) (Dai et al., 2021) specified by the Human Proteome Organization Proteomics Standards Initiative (HUPO-PSI) (Orchard et al., 2003). The JPDM format incorporates information corresponding to both SDRF and the Investigation Description Format (IDF), allowing experimental design, sample characteristics, and data-acquisition conditions to be handled together. In addition, JPDM records injection-replicate relationships, which are not explicitly represented in the standard SDRF specification. The use of injection-replicate information for the jPOST reanalysis can strengthen the validity for peptides that would otherwise be supported only weakly in an individual LC/MS/MS run.

The reanalysis results are incorporated into jPOSTdb so that they can be searched, compared, and reused by a broad range of users. Metadata generated or curated during reanalysis are retained as attributes of the corresponding data, making them directly available for database searches and downstream analyses. To improve search efficiency and ensure that datasets within jPOSTdb have internally consistent attributes, we established explicit rules for reorganizing deposited projects. An original repository project is first divided into one or more *reanalysis projects*, each consisting of raw-data files for which all dataset-defining metadata are identical, apart from replicate identifiers. Within each reanalysis project, files that differ only in a replicate identifier—excluding injection-replicate information—are assigned to separate *datasets*. Thus, if three replicate identifiers (other than injection-replicate identifiers) occur within a reanalysis project, the project is considered to contain three datasets, and these units are retained as the datasets presented in jPOSTdb. Importantly, a replicate identifier denotes the identity of a replicate set rather than the total number of replicates. For example, in an experiment involving biological samples from three patients, data derived from patients 1, 2, and 3 are assigned biological-replicate identifiers 1, 2, and 3, respectively.

Proteomics data reanalysis in jPOST

An initial version of the jPOST reanalysis workflow and its associated scripts was developed by 2024 and reported previously (Okuda et al., 2025). However, substantially increasing both the number of reanalyzed projects and the volume of data incorporated into jPOSTdb required a more efficient workflow with fewer manual intervention points. We therefore conducted a comprehensive review of the workflow and its implementation. The original developer,

one of the authors, first documented the purpose, assumptions, and intended behavior of each script. These descriptions and the complete set of scripts were then examined using Claude (Fable 5.1), a large language model, to identify inconsistencies, redundancies, and potentially inefficient procedures. The developer subsequently evaluated these observations, added technical explanations, and, where appropriate, challenged the model's interpretations. This iterative human–LLM review helped reconstruct the overall logic of the existing system and identify components that required redesign.

The jPOST reanalysis system comprises three operational components: reanalysis computation, reanalysis result deposition in the repository, and registration of the processed results in jPOSTdb. These components were designed to proceed with a degree of independence—particularly the repository and reanalysis operations—to avoid unnecessary delays. Such operational separation is also necessary because the three components are managed at geographically distributed sites in Niigata, Kyoto, and Kashiwanoha. However, this architecture created difficulties in transferring and reconciling identifiers and metadata among the components. In the previous workflow, metadata were generated in several stages: an initial metadata set was used to perform the reanalysis, after which additional metadata were derived from the analysis results. During the present redesign, explicit operational rules were introduced so that, wherever possible, the complete metadata set could be generated before reanalysis. This change simplified the exchange of identifiers and metadata among the three components and reduced dependencies between successive processing steps.

These improvements primarily concern operational standardization and scalability rather than the development of a new analytical method. Nevertheless, such workflow engineering is indispensable for the sustained expansion of a secondary proteomics database. Once routine operation of the redesigned workflow has been stabilized, we plan to use it to incorporate projects on a scale far exceeding 1,000 studies, thereby expanding the coverage and practical utility of jPOSTdb for large-scale comparative, metaproteomic, and trans-omics analyses.

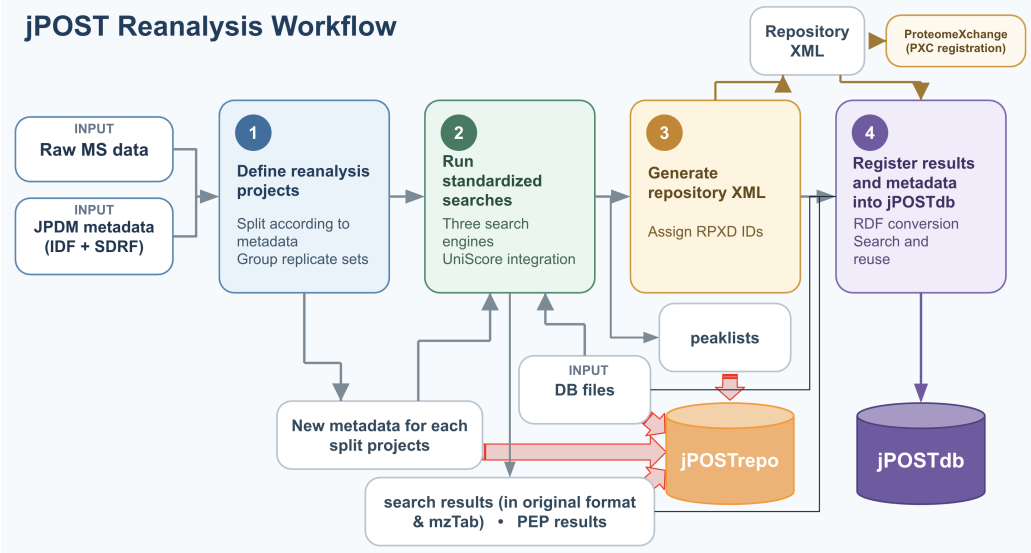


Figure 1: jPOST data reanalysis workflow

Metaproteome data visualization in jPOST

We developed and integrated a new set of visualizations that place metaproteome identifications in the context of metabolic pathways and the taxa contributing to them. The visualizations take as input the results of MetaPilot (Cheng & Figeys, 2026) (Cheng et al., 2020), a metaproteome analysis software, in which PSMs are used to infer proteins from a sequence database derived

from metagenome-assembled genomes (MAGs) (Setubal, 2021). The MAG protein FASTA dataset has been functionally annotated using eggNOG-mapper ver.2 (Cantalapiedra et al., 2021), and for each protein, fields from that output—such as the "EC number (Bairoch, 2000)," "COG classification (Galperin et al., 2025)," and "eggNOG taxonomy (Hernández-Plaza et al., 2026)" indicating the source organism's phylum—are utilized.

Pathway mapping. Proteins are linked to UniProt Pathway (The UniProt Consortium, 2025) annotations primarily through EC numbers, with COG assignments used as a fallback for proteins that cannot be mapped via EC numbers. Pathways are organized into a three-level hierarchy of functional groups, classes, and individual pathways. All values are weighted by PSM count, so each pathway reflects its share of the spectral validity rather than the number of distinct proteins. The proportion of PSMs mapped to any pathway, and how much of it comes through EC numbers versus COGs, is reported alongside each view to make annotation coverage explicit.

Per-sample and group comparison views. The per-sample view displays the pathway class composition of each sample as a stacked bar, grouped by the sample metadata (Figure 2, top left). The group comparison view summarizes each group as the mean composition of its samples, so that functional profiles can be compared between groups at a glance (Figure 2, top right). Differences between groups are tested for each pathway class, with correction by multiple testing (Kruskal–Wallis tests across the groups, Benjamini–Hochberg tests across the categories), and the result is summarized in the view.

Sample-level interpretation by phylum. To relate functional profiles to the organisms that produce the detected proteins, each sample page includes a tree heatmap (Figure 2, bottom). Rows follow the pathway hierarchy and columns correspond to phyla. A bar for each row shows its share of all mapped PSMs in the sample, while each cell shows the share of that phylum's own mapped PSMs. Because cells are normalized within each phylum, columns can be read as phylum-specific functional profiles independent of phylum abundance. Column headers show each phylum's share of PSMs and its pathway mapping rate. The hierarchy is collapsed to functional groups by default and can be expanded down to individual pathways, keeping even hundreds of pathways readable with their full names. The underlying values are also available through an API, allowing the same summaries to be used programmatically. Phylum assignments are taken from the eggNOG-mapper lineage, which reflects the NCBI Taxonomy snapshot of the eggNOG database version used (eggNOG 5.0) (Huerta-Cepas et al., 2019) and therefore retains older phylum names (e.g., Firmicutes), whereas the original taxonomy of the MAGs provided by MicrobiomeDatahub (Mori et al., 2026) uses the updated nomenclature (e.g., Bacillota).This situation arises due to the analysis tool being used, so it may be resolved in the future.

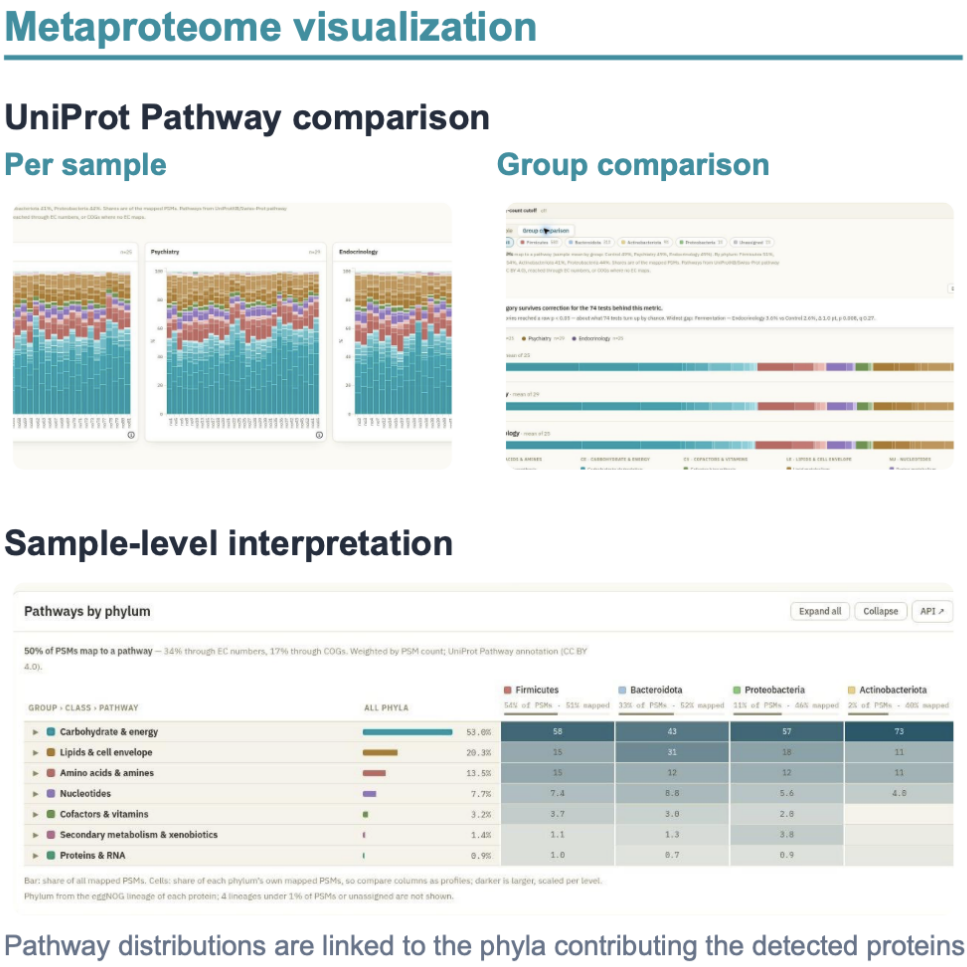


Figure 2: Metaproteome visualization. (Top left) Per-sample UniProt Pathway class composition, grouped by sample category. (Top right) Group comparison of mean pathway composition. (Bottom) Pathways by phylum for a single sample. Bars show the share of all mapped PSMs; cells show the share of each phylum's own mapped PSMs (darker is larger, scaled per level). Phylum assignments are taken from the eggNOG lineage of each protein. Pathway distributions are thus linked to the phyla contributing the detected proteins.

Enhancement of Mass++4 towards trans-omics research

Background: Development of Mass++4

Mass++4 is the latest version of the Mass++ software series (Tanaka et al., 2014), which has been downloaded approximately 20,000 times since the first version was released in June 2009. Mass++ is a data viewer designed for the interactive inspection of mass spectra and chromatograms. In contrast to the earlier versions, which were developed exclusively for Windows, Mass++4 was rewritten in Java to support multiple operating systems and provide a more sustainable basis for future development.

Up to version 2, Mass++ was developed primarily by a commercial company and was distributed free of charge as closed-source software. The source code of version 2.7.4 was first released. However, source code corresponding to functions covered by software patents acquired during the development of version 2 could not be included in the open-source release, designated

version 2.7.5. In addition, some external libraries and other dependencies could not be redistributed. These restrictions made continued maintenance and development of the original codebase impractical and prevented several analytical functions from being maintained. To overcome these limitations and extend support beyond Windows, we therefore undertook a complete reimplementation of Mass++ in Java. This redevelopment resulted in Mass++4; version 3 was intentionally omitted from the release sequence.

A manuscript of an academic paper describing Mass++4 is currently under revision (Tanaka et al., (Submitted)). In parallel with its preparation and revision, we implemented additional functions to connect Mass++4 with public data repositories, developed features in response to comments received during peer review, and expanded the documentation (a quick manual) available through GitHub. Repository connectivity is particularly important for the reuse of public MS datasets because it can shorten the path from data discovery and retrieval to direct visual inspection. The additional functions and documentation developed in this work are intended to improve the accessibility, reproducibility, and maintainability of Mass++4 as a general-purpose environment for examining MS-based omics data.

Implementation of a new function

Visual inspection of mass spectra and chromatograms remains essential in mass spectrometry research, including proteomics, metaproteomics, and metabolomics. Although these research fields employ substantially different data-processing and identification strategies, they share the need to examine the underlying spectral and chromatographic validity. Mass++4 addresses this common requirement and aims to serve as an indispensable companion for MS-based omics research. The functionality implemented here is designed to automate the simple yet cumbersome tasks that users typically perform during visual inspections. In such a workflow, a user would normally need to access a repository, select the appropriate data based on its metadata, download that data, and then open it in a viewer. The newly developed feature streamlines this process by: (i) Automatically accessing the data repository and retrieving metadata for all available datasets; (ii) Presenting the retrieved metadata in a list; (iii) Automatically downloading the corresponding dataset once the user selects an item from the list; and (iv) Automatically displaying the downloaded data. Consequently, the only action required of the user is to select the desired metadata from the list. While this functionality was previously implemented for MB-POST (Takahashi et al., 2026)—a repository for metabolomics data, constructed under the Shin-MassBank project—it has now been extended to support jPOSTrepo, a repository for proteomics data, during this BioHackathon. We plan to expand support to additional repositories in the future. Such functions also make it relevant to trans-omics studies integrating proteomic and metabolomic measurements. Mass++4 does not itself perform the complete computational integration of different omics layers; rather, it provides a shared environment in which researchers can inspect and validate the MS validity underlying protein- and metabolite-level results. By supporting access to heterogeneous public datasets and enabling their examination through a common interface, Mass++4 can provide a practical bridge between proteomics and metabolomics and contribute to the development of reliable trans-omics workflows.

Improving data interoperability between MB-POST and MetabolomicsHub

Background: Metadata standardization in metabolomics repositories

As in proteomics, the reliable reuse and reanalysis of mass spectrometry-based metabolomics data requires the preservation of raw data together with sufficiently detailed metadata. Without information describing the biological source, sample characteristics, experimental design, and analytical conditions, even the organism from which a dataset was obtained may be unclear,

potentially leading to inappropriate comparisons or integration of biologically incompatible data. Metadata are therefore indispensable not only for interpreting individual datasets but also for exchanging data among repositories and integrating metabolomics with other omics layers.

The importance of describing the relationship between samples and raw-data files was first formalized in the microarray field through the Minimum Information About a Microarray Experiment (MIAME) guidelines (Brazma et al., 2001) and the MAGE-TAB framework (Rayner et al., 2006). MAGE-TAB separates study-level information, represented in the Investigation Description Format (IDF), from sample-to-data relationships, represented in the Sample and Data Relationship Format (SDRF). This distinction is particularly important for MS-based omics because the spectra themselves constitute primary experimental evidence and must be retained for subsequent reanalysis. The proteomics community subsequently adapted these principles through the HUPO Proteomics Standards Initiative (HUPO-PSI), which developed the Minimum Information About a Proteomics Experiment (MIAPE) guidelines (Taylor et al., 2007) and related data standards. In parallel, ProteomeXchange established a framework through which participating proteomics repositories exchange datasets and operate according to shared submission and dissemination guidelines (Hermjakob & Apweiler, 2006).

Comparable initiatives were launched in metabolomics, including the Metabolomics Standards Initiative (MSI) (Fiehn et al., 2007) (Sansone et al., 2007) and MetabolomeXchange (Haug et al., 2017). However, these efforts did not result in a continuously maintained, community-wide minimum-information standard comparable to MIAME or MIAPE. Consequently, metabolomics repositories have developed their metadata models largely independently, reflecting their respective objectives, submission systems, and organizational principles. This heterogeneity limits cross-repository discovery and comparison and creates a substantial obstacle to large-scale data reuse. The problem is becoming increasingly important as metabolomics data are incorporated into trans-omics analyses and used to train or support artificial-intelligence systems, both of which require metadata that are consistent, machine-readable, and semantically interoperable.

Background: MetabolomicsHub and emerging international standards

Recent developments have provided a renewed basis for international standardization. Although HUPO-PSI was originally established for proteomics, some of its standards describe generic MS data and can therefore be applied across MS-based disciplines. For example, mzML (Deutsch, 2008) represents spectra and chromatograms independently of whether the underlying experiment is proteomic or metabolomic. PSI also extended mzTab (Griss et al., 2014), a tabular format for reporting identification and quantification results, to support metabolomics through mzTab-M (Hoffmann et al., 2019), also referred to within the mzTab specification as version 2.0. This development created a practical route for extending expertise and infrastructure originally developed in proteomics to metabolomics.

Building on this momentum, members of the metabolomics repository community established MetabolomicsHub in 2025 with support from the Chan Zuckerberg Biohub. MetabolomicsHub (Thakur et al., 2026) aims to provide a common framework for discovering and comparing datasets across major repositories as an alternative to the unsuccessful MSI. Together with mzTab-M, its common data model represents an important foundation for improving the standardization and interoperability of metabolomics data. Alignment with these international initiatives is therefore essential for repositories seeking to support cross-repository data exchange, large-scale reanalysis, and integration with proteomic and other omics data.

During BioHackathon 2026, we investigated how the data model of MB-POST, a metabolomics repository formally released in 2025, could be aligned with the Metabolomics Hub common data model and mzTab-M. Our immediate objective was not to replace the native MB-POST model, but to identify the transformations, vocabulary mappings, and metadata improvements required to make MB-POST datasets interoperable with these emerging standards.

Mapping MB-POST to the MetabolomicsHub common data model

MB-POST was developed with international collaboration in mind and intends to participate in Metabolomics Hub. To achieve this, MB-POST metadata must be representable using the Metabolomics Hub common data model. Integration frameworks have already been developed for MetaboLights (Yurekten et al., 2024), Metabolomics Workbench (Sud et al., 2016), and Global Natural Products Social Molecular Networking (GNPS) (Wang et al., 2016), allowing metadata from these repositories to be transformed into the common model and compared across repository boundaries. MB-POST will require an equivalent integration framework.

A central function of these frameworks is the mapping of repository-specific terms and controlled vocabularies to concepts used by the Metabolomics Hub model. We therefore performed a comprehensive survey of the metadata terms and values used across MB-POST. The resulting metadata profiles were compiled in the pymbpost resource, and the diversity of values recorded for each metadata field was examined. Metabolomics Hub represents metadata as a graph whose nodes and relationships are annotated with controlled-vocabulary terms. The next step is therefore to determine which MB-POST fields contain the concepts required by the common model and to map every relevant MB-POST value to the corresponding controlled-vocabulary term. Fields containing many distinct values will require substantial normalization and entity resolution before reliable automated conversion can be achieved.

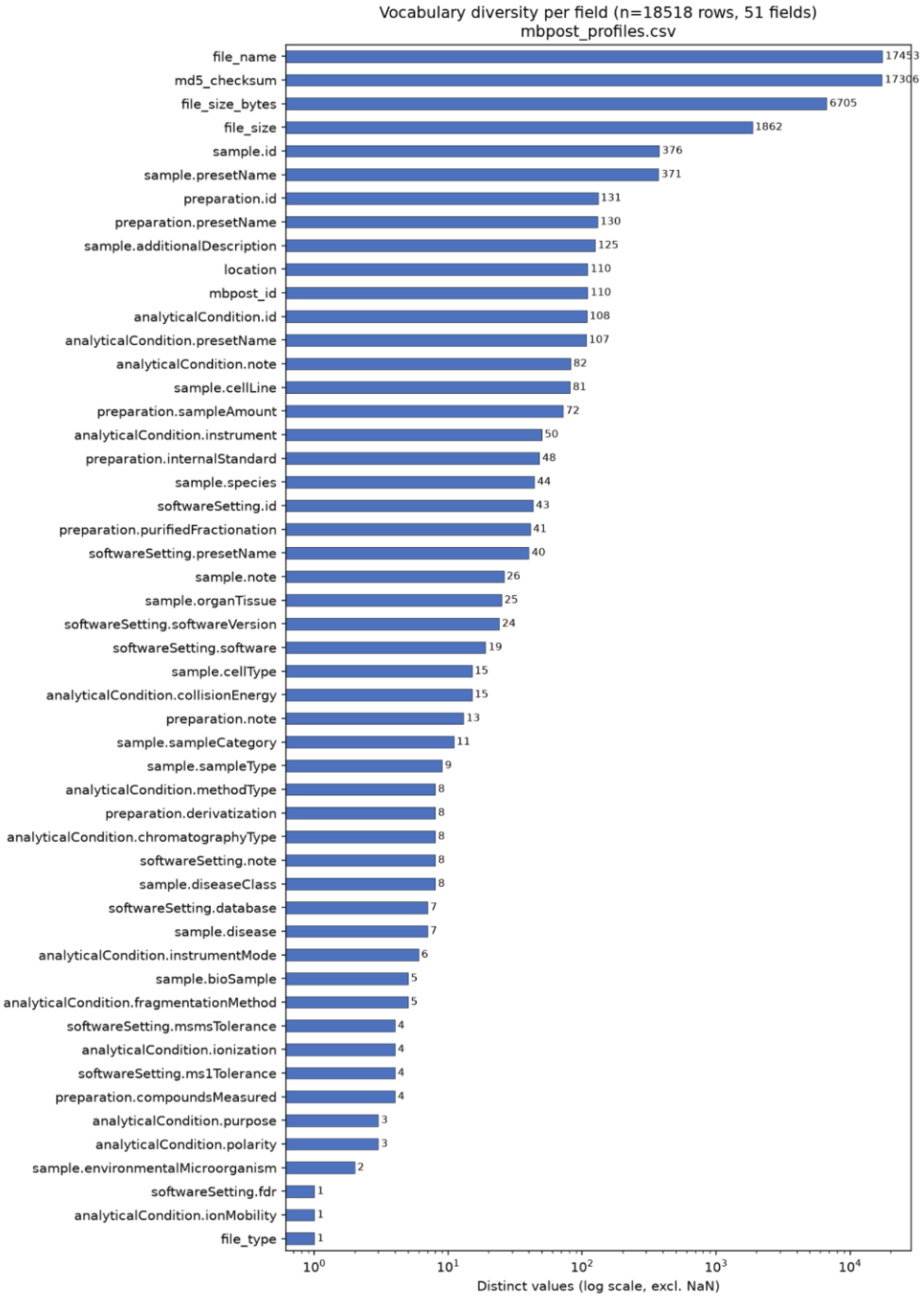


Figure 3: Numbers of distinct values recorded for metadata fields across all raw-data files registered in MB-POST.

Mapping MB-POST to mzTab-M

We also examined the compatibility of MB-POST with mzTab-M. The information required for mzTab-M can be broadly divided into two components: the definition of sample groups and experimental design, and the quantitative metabolite measurements obtained for individual samples. Because MB-POST does not currently impose a sufficiently strict and uniform format on submitted quantitative-result files, the second component cannot yet be converted

automatically. Standardizing these result tables, or requiring submission in a machine-readable format with explicitly defined metabolite and sample identifiers, will therefore be necessary for automated mzTab-M generation.

The BioHackathon investigation consequently focused on the first component: automated reconstruction of sample groups and experimental design from MB-POST metadata. The repository-wide vocabulary profiles showed that sample category information could generally be retrieved automatically. In contrast, the experimental *factor*—in the sense used by ISA-Tab (Johnson et al., 2021) to represent the variable deliberately changed or compared within a study—could be recorded only in the MB-POST "Additional information" field. The purpose of this field is not sufficiently explicit to submitters, and it has been populated in only a small proportion of existing datasets. As a result, experimental factors cannot currently be extracted reliably across MB-POST studies.

These findings identify two complementary requirements for improving MB-POST interoperability. First, existing repository-specific terms must be normalized and mapped to the controlled vocabularies used by Metabolomics Hub. Second, the MB-POST submission model should more explicitly capture experimental factors, sample-group definitions, and standardized quantitative results. Addressing these issues will enable the development of an MB-POST–to–Metabolomics Hub integration framework and facilitate mzTab-M export. More broadly, the resulting semantic and structural interoperability will be essential for cross-repository searches, automated reanalysis, AI-assisted data use, and trans-omics studies linking metabolomic measurements with proteomic and metaproteomic data.

Conclusion

In this study, we advanced several complementary components of an infrastructure for the reuse and integration of public MS-based omics data. First, we redesigned the jPOST proteomics reanalysis workflow to improve scalability, reduce manual intervention, and simplify the exchange of identifiers and metadata among the repository, reanalysis, and database components. The workflow combines multiple search engines using UniScore, incorporates detailed JPDM metadata, and reorganizes deposited data into internally consistent reanalysis projects and datasets. These improvements will support the systematic expansion of jPOSTdb while maintaining consistent analytical and metadata standards across studies.

Second, we developed visualization functions that make reanalyzed metaproteomic data more readily interpretable. By linking spectral evidence to metabolic pathways and microbial phyla, the new views enable users to compare functional profiles among samples and experimental groups and to examine which microbial groups contribute to particular biological processes. We also extended Mass++4 as a cross-platform environment for inspecting spectra and chromatograms and improved its connectivity with public repositories. Although Mass++4 does not itself integrate different omics layers computationally, it provides a common interface for examining the primary MS evidence underlying proteomic, metaproteomic, and metabolomic results. Together, these developments connect standardized processing with both biological interpretation and direct inspection of the underlying experimental data.

Third, we investigated the interoperability of MB-POST with the Metabolomics Hub common data model and mzTab-M. The analysis identified the need to map repository-specific terms to controlled vocabularies and to represent experimental factors, sample groups, and quantitative results more explicitly and consistently. It also showed that current MB-POST metadata can support partial automated conversion, while incomplete descriptions of experimental factors and heterogeneous quantitative-result formats remain important barriers. These findings provide a practical basis for developing an MB-POST integration framework for Metabolomics Hub and for enabling automated mzTab-M export.

Taken together, the work conducted during BioHackathon 2026 demonstrates that trans-omics research depends not only on new integration algorithms but also on coordinated improvements

throughout the data lifecycle: preservation of raw data, standardized metadata, reproducible reanalysis, controlled error estimation, interoperable data models, and visualization of both processed results and primary evidence. The developments reported here do not yet constitute a complete trans-omics platform. However, they establish essential connections between proteomics, metaproteomics, and metabolomics resources and clarify the remaining technical and semantic challenges. Future work will focus on operating the redesigned jPOST workflow at substantially larger scale, expanding pathway- and taxonomy-aware visualization, completing the mapping of MB-POST metadata to international controlled vocabularies, and standardizing quantitative metabolomics results. These efforts should ultimately enable public MS datasets to be discovered, reanalyzed, compared, and integrated across omics layers, providing a more reliable foundation for mechanistic investigation of biological systems.

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Resource

- Mass++4 source code repository:
  - https://github.com/masspp/mspp4-desktop/
- Mass++4 quick manual:
  - https://github.com/masspp/quick_manual
- pymbpost resource:
  - https://github.com/kozo2/pymbpost/blob/main/mbpost_profiles.zip

Author contributions

Akiyasu C. Yoshizawa: Conceptualization, Writing – original draft; Yuki Moriya: Software, Visualization, Writing – original draft; Kozo Nishida: Software, Investigation, Writing – original draft; Yushi Takahashi: Software, Investigation, Methodology; Satoshi Tanaka: Software, Investigation; Shin Kawano: Software, Conceptualization, Supervision; Susumu Goto: Conceptualization, Supervision, Project administration, Writing – original draft.

References

Bairoch, A. (2000). The enzyme database in 2000. *Nucleic Acids Research*, *28*(1), 304–305. https://doi.org/10.1093/nar/28.1.304

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *J Royal Stat Soc B*, *57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Brazma, A., Hingamp, P., Quackenbush, J., Sherlock, G., Spellman, P., Stoeckert, C., Aach, J., Ansorge, W., Ball, C. A., Causton, H. C., Gaasterland, T., Glenisson, P., Holstege, F. C., Kim, I. F., Markowitz, V., Matese, J. C., Parkinson, H., Robinson, A., Sarkans, U., … Vingron, M. (2001). Minimum information about a microarray experiment (miame)-toward standards for microarray data. *Nat Genet*, *29*(4), 365–71. https://doi.org/10.1038/ng1201-365

Burger, T. (2018). Gentle introduction to the statistical foundations of false discovery rate in quantitative proteomics. *J Proteome Res*, *17*(1), 12–22. https://doi.org/10.1021/acs.jproteome.7b00170

Cantalapiedra, C. P., Hernández-Plaza, A., Letunic, I., Bork, P., & Huerta-Cepas, J. (2021). Eggnog-mapper v2: Functional annotation, orthology assignments, and domain prediction at the metagenomic scale. *Molecular Biology and Evolution*, *38*(12), 5825–5829. https://doi.org/10.1093/molbev/msab293

Cheng, K., & Figeys, D. (2026). Metapilot enables adaptive genome-aware dda and dia metaproteomics using microbial genome catalogues. *bioRxiv*, 2026.06.12.728088. https://doi.org/10.64898/2026.06.12.728088

Cheng, K., Ning, Z., Zhang, X., Li, L., Liao, B., Mayne, J., & Figeys, D. (2020). Metalab 2.0 enables accurate post-translational modifications profiling in metaproteomics. *Journal of the American Society for Mass Spectrometry*, *31*(7), 1473–1482. https://doi.org/10.1021/jasms.0c00083

Cox, J., & Mann, M. (2008). Maxquant enables high peptide identification rates, individualized p.p.b.-range mass accuracies and proteome-wide protein quantification. *Nat Biotechnol*, *26*(12), 1367–1372. https://doi.org/10.1038/nbt.1511

Cox, J., Neuhauser, N., Michalski, A., Scheltema, R. A., Olsen, J. V., & Mann, M. (2011). Andromeda: A peptide search engine integrated into the maxquant environment. *Journal of Proteome Research*, *10*(4), 1794–1805. https://doi.org/10.1021/pr101065j

Dai, C., Fullgrabe, A., Pfeuffer, J., Solovyeva, E. M., Deng, J., Moreno, P., Kamatchinathan, S., Kundu, D. J., George, N., Fexova, S., Gruning, B., Foll, M. C., Griss, J., Vaudel, M., Audain, E., Locard-Paulet, M., Turewicz, M., Eisenacher, M., Uszkoreit, J., … Perez-Riverol, Y. (2021). A proteomics sample metadata representation for multiomics integration and big data analysis. *Nat Commun*, *12*(1), 5854. https://doi.org/10.1038/s41467-021-26111-3

Deutsch, E. (2008). Mzml: A single, unifying data format for mass spectrometer output. *Proteomics*, *8*(14), 2776–7. https://doi.org/10.1002/pmic.200890049

Elias, J. E., & Gygi, S. P. (2007). Target-decoy search strategy for increased confidence in large-scale protein identifications by mass spectrometry. *Nat Methods*, *4*(3), 207–14. https://doi.org/10.1038/nmeth1019

Fiehn, O., Robertson, D., Griffin, J., van der Werf, M., Nikolau, B., Morrison, N., Sumner, L. W., Goodacre, R., Hardy, N. W., Taylor, C., Fostel, J., Kristal, B., Kaddurah-Daouk, R., Mendes, P., van Ommen, B., Lindon, J. C., & Sansone, S.-A. (2007). The metabolomics standards initiative (msi). *Metabolomics*, *3*(3), 175–178. https://doi.org/10.1007/s11306-007-0070-6

Galperin, M. Y., Vera Alvarez, R., Karamycheva, S., Makarova, K. S., Wolf, Y. I., Landsman, D., & Koonin, E. V. (2025). Cog database update 2024. *Nucleic Acids Research*, *53*(D1), D356–D363. https://doi.org/10.1093/nar/gkae983

Griss, J., Jones, A. R., Sachsenberg, T., Walzer, M., Gatto, L., Hartler, J., Thallinger, G. G., Salek, R. M., Steinbeck, C., Neuhauser, N., Cox, J., Neumann, S., Fan, J., Reisinger, F., Xu, Q.-W., del Toro, N., Pérez-Riverol, Y., Ghali, F., Bandeira, N., … Hermjakob, H. (2014). The mztab data exchange format: Communicating mass-spectrometry-based proteomics and metabolomics experimental results to a wider audience*. *Molecular & Cellular Proteomics*, *13*(10), 2765–2775. https://doi.org/https://doi.org/10.1074/mcp.O113.036681

Haug, K., Salek, R. M., & Steinbeck, C. (2017). Global open data management in metabolomics. *Current Opinion in Chemical Biology*, *36*, 58–63. https://doi.org/https://doi.org/10.1016/j.cbpa.2016.12.024

Hermjakob, H., & Apweiler, R. (2006). The proteomics identifications database (pride) and the proteomexchange consortium: Making proteomics data accessible. *Expert Rev Proteomics*, *3*(1), 1–3. https://doi.org/10.1586/14789450.3.1.1

Hernández-Plaza, A., Deng, Z., Robledo-Yagüe, F., Szklarczyk, D., von Mering, C., Bork, P., & Huerta-Cepas, J. (2026). Eggnog v7: Phylogeny-based orthology predictions and functional annotations. *Nucleic Acids Research*, *54*(D1), D402–D408. https://doi.org/10.1093/nar/gkaf1249

Hoffmann, N., Rein, J., Sachsenberg, T., Hartler, J., Haug, K., Mayer, G., Alka, O., Dayalan, S., Pearce, J. T. M., Rocca-Serra, P., Qi, D., Eisenacher, M., Perez-Riverol, Y., Vizcaíno, J. A., Salek, R. M., Neumann, S., & Jones, A. R. (2019). Mztab-m: A data standard for sharing quantitative results in mass spectrometry metabolomics. *Anal Chem*, *91*(5), 3302–3310. https://doi.org/10.1021/acs.analchem.8b04310

Huerta-Cepas, J., Szklarczyk, D., Heller, D., Hernández-Plaza, A., Forslund, S. K., Cook, H., Mende, D. R., Letunic, I., Rattei, T., Jensen, L. J., von Mering, C., & Bork, P. (2019). Eggnog 5.0: A hierarchical, functionally and phylogenetically annotated orthology resource based on 5090 organisms and 2502 viruses. *Nucleic Acids Research*, *47*(D1), D309–D314. https://doi.org/10.1093/nar/gky1085

Ishihama, Y. (2019). From bench to internet: Sharing proteomics data and methods through the open access journal. *Journal of Proteome Data and Methods*, *1*, 1–1. https://doi.org/10.14889/jpdm.2019.0001

Johnson, D., Batista, D., Cochrane, K., Davey, R. P., Etuk, A., Gonzalez-Beltran, A., Haug, K., Izzo, M., Larralde, M., Lawson, T. N., Minotto, A., Moreno, P., Nainala, V. C., O'Donovan, C.,

Pireddu, L., Roger, P., Shaw, F., Steinbeck, C., Weber, R. J. M., … Rocca-Serra, P. (2021). Isa api: An open platform for interoperable life science experimental metadata. *Gigascience*, *10*(9). https://doi.org/10.1093/gigascience/giab060

Käll, L., Storey, J. D., MacCoss, M. J., & Noble, W. S. (2008). Assigning significance to peptides identified by tandem mass spectrometry using decoy databases. *J Proteome Res*, *7*(1), 29–34. https://doi.org/10.1021/pr700600n

Kell, D. B., & Oliver, S. G. (2016). The metabolome 18 years on: A concept comes of age. *Metabolomics*, *12*(9), 148. https://doi.org/10.1007/s11306-016-1108-4

Kong, A. T., Leprevost, F. V., Avtonomov, D. M., Mellacheruvu, D., & Nesvizhskii, A. I. (2017). Msfragger: Ultrafast and comprehensive peptide identification in mass spectrometry-based proteomics. *Nat Methods*, *14*(5), 513–520. https://doi.org/10.1038/nmeth.4256

Lottspeich, F. (2018). Proteomics—an unexpected journey into the complexity of protein structures and functions. *EuPA Open Proteomics*, *21*, 1–2. https://doi.org/https://doi.org/10.1016/j.euprot.2018.11.001

Mori, H., Fujisawa, T., Higashi, K., Tanizawa, Y., Nakagawa, Z., Nishide, H., Fujiyoshi, M., Nakamura, Y., Uchiyama, I., Matsui, M., & Yamada, T. (2026). Microbiome datahub: An open-access platform integrating environmental metadata, taxonomy, and functional annotation for comprehensive metagenome-assembled genome datasets. *Microbiome*, *14*(1), 119. https://doi.org/10.1186/s40168-026-02385-x

Moriya, Y., Kawano, S., Okuda, S., Watanabe, Y., Matsumoto, M., Takami, T., Kobayashi, D., Yamanouchi, Y., Araki, N., Yoshizawa, A. C., Tabata, T., Iwasaki, M., Sugiyama, N., Tanaka, S., Goto, S., & Ishihama, Y. (2019). The jpost environment: An integrated proteomics data repository and database. *Nucleic Acids Res*, *47*(D1), D1218–D1224. https://doi.org/10.1093/nar/gky899

Okuda, S., Watanabe, Y., Moriya, Y., Kawano, S., Yamamoto, T., Matsumoto, M., Takami, T., Kobayashi, D., Araki, N., Yoshizawa, A. C., Tabata, T., Sugiyama, N., Goto, S., & Ishihama, Y. (2017). Jpostrepo: An international standard data repository for proteomes. *Nucleic Acids Res*, *45*(D1), D1107–D1111. https://doi.org/10.1093/nar/gkw1080

Okuda, S., Yoshizawa, A. C., Kobayashi, D., Takahashi, Y., Watanabe, Y., Moriya, Y., Hatano, A., Takami, T., Matsumoto, M., Araki, N., Tabata, T., Iwasaki, M., Sugiyama, N., Kodera, Y., Tanaka, S., Goto, S., Kawano, S., & Ishihama, Y. (2025). Jpost environment accelerates the reuse and reanalysis of public proteome mass spectrometry data. *Nucleic Acids Res*, *53*(D1), D462–D467. https://doi.org/10.1093/nar/gkae1032

Oliver, S. G., Winson, M. K., Kell, D. B., & Baganz, F. (1998). Systematic functional analysis of the yeast genome. *Trends in Biotechnology*, *16*(9), 373–378. https://doi.org/10.1016/S0167-7799(98)01214-1

Orchard, S., Hermjakob, H., & Apweiler, R. (2003). The proteomics standards initiative. *Proteomics*, *3*(7), 1374–6. https://doi.org/10.1002/pmic.200300496

Perkins, D. N., Pappin, D. J. C., Creasy, D. M., & Cottrell, J. S. (1999). Probability-based protein identification by searching sequence databases using mass spectrometry data. *ELECTROPHORESIS*, *20*(18), 3551–3567. https://doi.org/10.1002/(SICI)1522-2683(19991201)20:18<3551::AID-ELPS3551>3.0.CO;2-2

Rayner, T. F., Rocca-Serra, P., Spellman, P. T., Causton, H. C., Farne, A., Holloway, E., Irizarry, R. A., Liu, J., Maier, D. S., Miller, M., Petersen, K., Quackenbush, J., Sherlock, G., Stoeckert, J., C. J., White, J., Whetzel, P. L., Wymore, F., Parkinson, H., Sarkans, U., … Brazma, A. (2006). A simple spreadsheet-based, miame-supportive format for microarray data: Mage-tab. *BMC Bioinformatics*, *7*, 489. https://doi.org/10.1186/1471-2105-7-489

Sansone, S. A., Fan, T., Goodacre, R., Griffin, J. L., Hardy, N. W., Kaddurah-Daouk, R., Kristal, B. S., Lindon, J., Mendes, P., Morrison, N., Nikolau, B., Robertson, D., Sumner, L. W., Taylor, C., van der Werf, M., van Ommen, B., Fiehn, O., & Members, M. S. I. B. (2007). The metabolomics standards initiative. *Nat Biotechnol*, *25*(8), 846–8. https://doi.org/10.1038/nbt0807-846b

Setubal, J. C. (2021). Metagenome-assembled genomes: Concepts, analogies, and challenges. *Biophysical Reviews*, *13*(6), 905–909. https://doi.org/10.1007/s12551-021-00865-y

Shuken, S. R. (2023). An introduction to mass spectrometry-based proteomics. *Journal of Proteome Research*, *22*(7), 2151–2171. https://doi.org/10.1021/acs.jproteome.2c00838

Smith, R., Mathis, A. D., Ventura, D., & Prince, J. T. (2014). Proteomics, lipidomics, metabolomics: A mass spectrometry tutorial from a computer scientist's point of view. *BMC Bioinformatics*, *15*(7), S9. https://doi.org/10.1186/1471-2105-15-S7-S9

Sud, M., Fahy, E., Cotter, D., Azam, K., Vadivelu, I., Burant, C., Edison, A., Fiehn, O., Higashi, R., Nair, K. S., Sumner, S., & Subramaniam, S. (2016). Metabolomics workbench: An international repository for metabolomics data and metadata, metabolite standards, protocols, tutorials and training, and analysis tools. *Nucleic Acids Res*, *44*(D1), D463–70. https://doi.org/10.1093/nar/gkv1042

Tabata, T., Yoshizawa, A. C., Ogata, K., Chang, C.-H., Araki, N., Sugiyama, N., & Ishihama, Y. (2025). Uniscore, a unified and universal measure for peptide identification by multiple search engines. *Mol Cell Proteomics*, *24*(7), 101010. https://doi.org/10.1016/j.mcpro.2025.101010

Takahashi, Y., Yoshizawa, A. C., Kiuchi, T., Hayasaka, R., Torigoe, T., Takahashi, M., Oka, T., Matsuzawa, Y., Ogawa, M., Izumi, Y., Tsugawa, H., Hirayama, A., Matsuda, F., & Okuda, S. (2026). Reviving data potential—the reanalysis-driven metabolomics data repository mb-post. *Analytical Chemistry*, *98*(31), 22609–22617. https://doi.org/10.1021/acs.analchem.6c00697

Tanaka, S., Fujita, Y., Parry, H. E., Yoshizawa, A. C., Morimoto, K., Murase, M., Yamada, Y., Yao, J., Utsunomiya, S. I., Kajihara, S., Fukuda, M., Ikawa, M., Tabata, T., Takahashi, K., Aoshima, K., Nihei, Y., Nishioka, T., Oda, Y., & Tanaka, K. (2014). Mass++: A visualization and analysis tool for mass spectrometry. *J Proteome Res*, *13*(8), 3846–3853. https://doi.org/10.1021/pr500155z

Tanaka, S., Murase, M., Kato, M., Yamamoto, H., Matsubara, M., Takahashi, Y., Tabata, T., Kusano, M., Kawano, S., Okuda, S., & Yoshizawa, A. C. ((Submitted)).

Taylor, C. F., Paton, N. W., Lilley, K. S., Binz, P. A., Julian, J., R. K., Jones, A. R., Zhu, W., Apweiler, R., Aebersold, R., Deutsch, E. W., Dunn, M. J., Heck, A. J., Leitner, A., Macht, M., Mann, M., Martens, L., Neubert, T. A., Patterson, S. D., Ping, P., … Hermjakob, H. (2007). The minimum information about a proteomics experiment (miape). *Nat Biotechnol*, *25*(8), 887–893. https://doi.org/10.1038/nbt1329

Thakur, M., Bosc, N., Brooksbank, C., Ernst, C., Freeberg, M. A., Gurwitz, K. T., Hermjakob, H., Hulcoop, D. G., Martin, M. J., McDonagh, E. M., Mithani, A., O'Boyle, N. M., Ochoa, D., Payne, T., Perez-Riverol, Y., Sarkans, U., Sokolov, A., Staudt, N., Stephenson, J. D., … McEntyre, J. (2026). Embl's european bioinformatics institute (embl-ebi) in 2025. *Nucleic Acids Research*, *54*(D1), D10–D19. https://doi.org/10.1093/nar/gkaf1078

The UniProt Consortium. (2025). UniProt: The universal protein knowledgebase in 2025. *Nucleic Acids Res*, *53*(D1), D609–d617. https://doi.org/10.1093/nar/gkae1010

Wang, M., Carver, J. J., Phelan, V. V., Sanchez, L. M., Garg, N., Peng, Y., Nguyen, D. D., Watrous, J., Kapono, C. A., Luzzatto-Knaan, T., Porto, C., Bouslimani, A., Melnik, A. V., Meehan, M. J., Liu, W. T., Crüsemann, M., Boudreau, P. D., Esquenazi, E., Sandoval-Calderón, M., … Phapale, P., et al. (2016). Sharing and community curation of mass spectrometry data with global natural products social molecular networking. *Nat Biotechnol*, *34*(8), 828–837. https://doi.org/10.1038/nbt.3597

Wilkins, M. R., Pasquali, C., Appel, R. D., Ou, K., Golaz, O., Sanchez, J.-C., Yan, J. X., Gooley, A. A., Hughes, G., Humphery-Smith, I., Williams, K. L., & Hochstrasser, D. F. (1996). From proteins to proteomes: Large scale protein identification by two-dimensional electrophoresis and arnino acid analysis. *Bio/Technology*, *14*(1), 61–65. https://doi.org/10.1038/nbt0196-61

Wilmes, P., & Bond, P. L. (2004). The application of two-dimensional polyacrylamide gel electrophoresis and downstream analyses to a mixed community of prokaryotic microorganisms. *Environmental Microbiology*, *6*(9), 911–920. https://doi.org/https://doi.org/10.1111/j.1462-2920.2004.00687.x

Wilmes, P., Heintz-Buschart, A., & Bond, P. L. (2015). A decade of metaproteomics: Where we stand and what the future holds. *PROTEOMICS*, *15*(20), 3409–3417. https://doi.org/https://doi.org/10.1002/pmic.201500183

Yugi, K., Kubota, H., Hatano, A., & Kuroda, S. (2016). Trans-omics: How to reconstruct biochemical networks across multiple 'omic' layers. *Trends in Biotechnology*, *34*(4), 276–290. https://doi.org/https://doi.org/10.1016/j.tibtech.2015.12.013

Yurekten, O., Payne, T., Tejera, N., Amaladoss, F. X., Martin, C., Williams, M., & O'Donovan, C. (2024). Metabolights: Open data repository for metabolomics. *Nucleic Acids Res*, *52*(D1), D640–d646. https://doi.org/10.1093/nar/gkad1045